A FUNCTOR COMMUTING WITH COUNTABLE PRODUCTS COMMUTES WITH DERIVED LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; each of its steps names the result it uses and the closing sentence follows the comparison displayed just before it.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Existence of K-injective resolutions. Same chapter, Theorem theorem-K-injective-embedding-grothendieck. Supplies that the derived functor exists because the source category has enough K-injectives.
- Products in the derived category. Same chapter, Lemma lemma-derived-products and Derived Categories, Lemma derived-lemma-products. Supply that products in the derived category are computed termwise on K-injective complexes, which is what allows the computation of the derived functor on products.
- Derived limits. Derived Categories, the description of derived limits as objects obtained from products of K-injective complexes. Supplies the description of the derived limit that the proof compares with the functor applied to it.

Source reference. Original theorem 079P: sources/stacks-injectives/pinned-original-source.tex, source lines 2024–2024. Navigation pointer invoked at proof line 2254; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070Y: sources/stacks-derived/pinned-original-source.tex, source lines 9964–9964. Navigation pointer invoked at proof line 2299;

Date: stacks-project:injectives.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 08TC: sources/stacks-derived/pinned-original-source.tex, source lines 10659–10659. Navigation pointer invoked at proof line 2302; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07D9: sources/stacks-injectives/pinned-original-source.tex, source lines 2226–2226. Navigation pointer invoked at proof line 2268; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07KC: sources/stacks-derived/pinned-original-source.tex, source lines 10692–10692. Navigation pointer invoked at proof line 2311; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor of abelian categories. Assume*

(1) *$\mathcal{A}$ is a Grothendieck abelian category,*
(2) *$\mathcal{B}$ has exact countable products, and*
(3) *F commutes with countable products.*

Then $RF : D(\mathcal{A}) \to D(\mathcal{B})$ commutes with derived limits.

Proof. Observe that RF exists as $\mathcal{A}$ has enough K-injectives (Theorem 079P and Derived Categories, Lemma 070Y). The statement means that if $K = R\lim K_n$, then $RF(K) = R\lim RF(K_n)$. See Derived Categories, Definition 08TC for notation. Since RF is an exact functor of triangulated categories it suffices to see that RF commutes with countable products of objects of $D(\mathcal{A})$. In the proof of Lemma 07D9 we have seen that products in $D(\mathcal{A})$ are computed by taking products of K-injective complexes and moreover that a product of K-injective complexes is K-injective. Moreover, in Derived Categories, Lemma 07KC we have seen that products in $D(\mathcal{B})$ are computed by taking termwise products. Since RF is computed by applying F to a K-injective representative and since we've assumed F commutes with countable products, the lemma follows. □